

Firm-level price wedges for pricewedge.com: design decisions and why they were made

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Why this document exists

The website reports a price wedge for an individual firm. The paper the site is built on (Binsbergen, Boons, Opp and Tamoni, *Dynamic Asset (Mis)Pricing: Build-up versus Resolution Anomalies*) estimates price wedges for decile portfolios and carries them to firms by a regression on characteristics. Rebuilding that construction from raw CRSP and Compustat data exposed choices inside it that the paper does not discuss and that move the number a firm is given by ten to twenty percentage points. Some of those choices were made one way in the paper and would be made another way if the objective is the best possible firm-level estimate rather than a replication.

This document records each such choice: what the options are, what evidence separates them, which was chosen for the site and why, and what would change the decision. It is meant to be read on its own; the supporting numbers are in `quality.pdf` (the mapping and level tests), `signals.pdf` (every signal's construction) and the scripts named in the appendix. Every number quoted here comes from a script that can be re-run.

The order follows the logic of the construction: what a wedge is, what a firm's wedge should mean, which signals define the portfolios, and which sample the wedges are estimated on.

1 The object: what a price wedge is, and the one calibration that defines it

A price wedge is minus the log of the expected discounted cash flow, over the fifteen years after formation, per dollar of price at formation:

$$\text{PW} = -\log \mathbb{E} \left[\frac{\sum_{s=1}^{180} m_{t,t+s} D_{t+s} + m_{t,t+180} P_{t+180}}{P_t} \right],$$

where m is a stochastic discount factor of the CAPM form and the expectation is the mean over formation cohorts (one cohort per month, July 1964 onward). A negative wedge means the price is below the discounted cash flows that followed: underpricing. The discount factor has one free parameter, the price of market risk λ , and the paper fixes it by one condition: **the wedge of the value-weighted market portfolio is zero**. Everything else is measured relative to that. A wedge is therefore a statement about a group of dollars relative to the average dollar in the market. This

calibration is the identifying normalisation of the whole method and is kept unchanged in every decision below.

Two properties of the object matter for what follows. First, a portfolio’s wedge is a dollar-weighted quantity: it is the wedge of one dollar spread across the portfolio’s members in proportion to their capitalisation, so it is set by the large members. Second, a wedge is imprecise: the 463 formation cohorts of the paper’s window overlap in 179 of their 180 months and are worth two or three independent observations, so a single portfolio wedge carries a standard error near 30 percentage points, almost all of it common to every portfolio (the market’s realised path), with about 3 to 4 points that is specific to the portfolio.

One realised number frames the rest of the document. With the paper’s λ , the value-weighted universe has a wedge of zero by construction. The *equal-weighted* universe, one dollar in every firm, has a realised wedge of -24.6 percentage points. The average firm is a small firm, and under this discount factor small firms are heavily underpriced. So “the average dollar” and “the average firm” are 25 points apart, and any firm-level number has to say which of the two it is about.

2 Decision A: what the firm-level number means (taken: candidate 3)

How a portfolio wedge is carried to a firm

The 57 characteristics give 570 decile portfolios with wedges. Each portfolio is described by a *profile*: its members’ percentile ranks on all 57 characteristics, averaged over the members and over time. A regression of the 570 wedges on the 570 profiles gives a rule from characteristics to wedge, and evaluating that rule at a firm’s own ranks gives the firm’s wedge. The paper does this with three principal components of the profiles of the 114 extreme deciles; the pipeline does it with a penalised regression on all 570, which is better on every portfolio-level criterion ([quality.pdf](#), first section). That choice is not the decision here. The decision is what “profile” means, and it was hidden.

A profile can describe a portfolio’s typical *member*: the equal-weighted mean of the members’ ranks. Or it can describe its typical *dollar*: the capitalisation-weighted mean. For the value-weighted book-to-market decile 10, the typical member has a size rank near 0.4 and the typical dollar a size rank near 0.98, because a few large firms hold most of the decile’s capitalisation. The paper’s code uses the equal-weighted mean (with the comment “SHOULD THIS BE VALUE WEIGHTED AVERAGE?????” beside it); the pipeline had used the capitalisation-weighted mean. This one choice moves every firm-level wedge by ten to twenty-seven points, and it is the whole reason the pipeline’s firm wedges sat eleven points below the published ones. With equal-weighted profiles the pipeline reproduces the published firm wedges to 1.2 points in level (slope 1.04, correlation 0.99); the residual is the paper’s log-bias correction on the portfolio wedges, which is about one point and enters once, on the left-hand side.

The test that decides between candidates

Correlation between two firm-level series says nothing about levels, and levels are what the site prints. The test used throughout is the *realised-wedge test*: treat a candidate firm wedge as a characteristic, sort firms into deciles on it every month at NYSE breakpoints, hold each decile fifteen years as an equal-weighted and as a value-weighted portfolio, compute the decile’s realised

wedge with the same discount factor, and regress it on the mean wedge the candidate assigned to the decile's members. A candidate whose levels can be trusted shows a slope of one and an intercept of zero. Equal-weighted deciles answer “do firms realise what they were told”, which is the website's question; value-weighted deciles answer “does the candidate respect the calibration”. A second check follows directly from the calibration: the capitalisation-weighted mean of all firm wedges must equal the market wedge, zero.

The three candidates

Candidate 1: the published construction. Value-weighted portfolio wedges regressed on equal-weighted profiles. The regression learns “a portfolio whose typical member is a small value firm has wedge W ”, but W was earned by the portfolio's large members. Evaluated at a small firm's own ranks, the small firm receives the large firm's wedge. The test shows it: on equal-weighted deciles the published series has slope 0.64 and intercept -18 points; the firms it calls fairly priced (decile 5, assigned -4) realise -25 ; only the most underpriced decile realises what it is assigned. Its firm-level average is -7 against a realised -22 for the same firms, and its capitalisation-weighted mean is $+9$ where the calibration requires zero. The sentence the site could claim under this candidate is “this is the number the JFE method assigns”, which is true and is not what firms like this one realised.

Candidate 2: per dollar. Value-weighted wedges regressed on value-weighted profiles. Now the regression is internally consistent: “a dollar in firms with size rank 0.98 and book-to-market rank 0.95 earned wedge W ”, and W belongs to those dollars. The difficulty is where the rule is evaluated. Nearly every portfolio's dollar-profile has a size rank above 0.85 and the site asks about firms with size rank 0.4, so the fitted line is extended far outside the points it was fitted on. The average is right (capitalisation-weighted mean $+1$, intercepts -3) and the extremes are exaggerated: the most underpriced decile is assigned about -70 and realises -42 ; the slopes are 0.67 equal-weighted and 0.79 value-weighted. Heavier shrinkage repairs the value-weighted slope (1.02 at a penalty of 1000) and not the equal-weighted one (0.79). The sentence: “a dollar invested in firms like this one was mispriced by X ”. Right for the market as a whole, exaggerated for small firms.

Candidate 3: per firm, with the aggregate pinned. Add to the regression the same 570 deciles held with *equal* weights, whose wedge is the wedge of the average member, each described by the equal-weighted profile of its members, alongside the value-weighted deciles with their value-weighted profiles. Both kinds of evidence are then in the fit: “the average small value firm realised -42 ” from the equal-weighted decile, next to “the average value dollar realised -32 ” from the value-weighted one. A small firm's number is interpolated from portfolios whose members resemble it, and the value-weighted portfolios keep the dollar-weighted total anchored. On the test: slope 0.86 to 0.88 and intercept within one point on equal-weighted deciles, mean absolute gap of 3.5 to 4.5 points across the ten deciles, capitalisation-weighted mean of all firm wedges 0.0 points. Out of sample (fitted on cohorts formed before 1988, evaluated on the later cohorts) the slopes hold at 0.87 to 0.99 and the level moves with the market, as every candidate's does. The sentence: “firms with this profile were mispriced by X on average, each firm counting once; dollar-weighted across all firms, mispricing is zero, as the calibration requires.”

The decision and its reasons

Candidate 3 is adopted for the site’s headline series. The reasons, in order of weight:

1. It is the only candidate whose number was checked against realised outcomes at the level of the firm and found right, equal-weighted and value-weighted at once.
2. Its sentence is the one a user looking up a single company expects the number to mean.
3. It keeps the paper’s calibration intact rather than working around it: the value-weighted portfolios remain in the fit and the aggregate check holds at 0.0.
4. It does not sacrifice ranking: all three candidates rank firms similarly (correlations 0.9 and above); the candidates differ in levels, and levels are the objective.

The published construction (candidate 1) is kept as a labelled reference series, because it is what the paper reports and readers will want to see the relation. Candidate 2 is not shown.

What this changes in the construction

- Equal-weighted decile portfolios are formed for every sort (`portfolio_wedges.py --weighting equal`) and priced at the λ of the value-weighted calibration; the calibration itself is not touched.
- Profiles are computed with the weighting of the portfolio they describe (`portfolio_profiles.py --weighting equal` for the equal-weighted set).
- The mapping is a penalised direct regression on the 1,140 portfolios with their matching profiles. The penalty is chosen by the leave-one-family-out fit and accepted only if the realised-level test passes (slope of at least 0.85 and intercept within about 3 points equal-weighted; capitalisation-weighted mean within a point or two of the market wedge).
- The realised-level test and the aggregate check become acceptance gates in the pipeline (`realised_levels.py`); no series goes to the site without them.

What would change the decision

Evidence that the equal-weighted portfolios’ wedges are too noisy to help (they are not: their idiosyncratic standard errors are of the same size as the value-weighted ones), or a decision that the site’s number should be a per-dollar statement. Neither is expected.

A consequence for the paper

The published firm-level wedges are candidate 1. Any use of them in levels inherits the -18 -point intercept: the average firm’s mispricing is understated by about that amount, and firm-level regressions that use the wedge as a regressor in levels (the investment- q analysis) are affected. Rankings are not. This is worth raising with the coauthors before the extension to more characteristics is built.

3 Decision B: which signal definitions the site’s portfolios are built on (taken: as the original papers define them)

Why it matters

The mapping learns from the wedges of the 570 (now 1,140) portfolios, and a portfolio is defined by how its sorting characteristic was built. The rebuild verified 56 of the 57 characteristics against the paper’s own panels, and in doing so found six places where what the paper did is not what the paper describes. For the site the question is not which is right in the replication sense (both are known now) but which the site should claim as the definition of its inputs.

The sources, signal by signal

The rule adopted for this decision is that each signal is defined by the paper the anomaly comes from, read through Freyberger, Neuhierl and Weber (2020), whose appendix is the list the paper takes its characteristics from and the source of Table A.1’s wording. What was checked, from the original texts where they could be obtained (15 September 2026):

Operating accruals (OA) and their absolute value (AOA). Sloan (1996) defines accruals as the change in non-cash current assets minus the change in current liabilities other than debt in current liabilities and taxes payable, *minus* depreciation, scaled by average total assets (the Chen–Zimmermann rendering of Sloan’s equation). FNW state the same components, scaled by lagged total assets. The paper’s data adds depreciation instead of subtracting it; the sort it produces agrees with the documented one at only 0.29. AOA is the absolute value of OA (Bandyopadhyay, Huang and Wirjanto 2010). **Settled: subtract depreciation, scale by lagged total assets as FNW do.** The scaling difference between Sloan and FNW is immaterial for a rank.

Standardised unexplained volume (SUV). Garfinkel (2009, equations 4–6) defines a *daily* measure: $SUV_{i,t} = UV_{i,t}/S_i$, where $UV_{i,t}$ is the day’s volume minus $\hat{\alpha}_i + \hat{\beta}_1|R_{i,t}|^+ + \hat{\beta}_2|R_{i,t}|^-$, the regression estimated over a control window that precedes the measurement day, and S_i the standard deviation of that regression’s residuals over the estimation window. In his event setting the measure is averaged over the event days. FNW turn it into a monthly characteristic with the regression estimated on the previous month and the unexplained volume taken in the current month, which is what Table A.1 says. Neither source singles out one day. The paper’s data is the last trading day’s residual from a regression run within the same month. **Settled: for every trading day of month t , the standardised residual from the regression estimated on the days of month $t - 1$; the characteristic is the mean over the days of month t .** The aggregation by the mean is the one choice not dictated by the sources; it is Garfinkel’s own aggregation over his event window.

Detrended turnover (DTO). Garfinkel (2009, equation 3) defines daily market-adjusted turnover as the firm’s volume over its shares minus the same ratio for all NYSE and AMEX stocks, and “change in turnover” as that quantity minus its *mean* over a control period (twenty trading days). FNW’s characteristic subtracts the daily market turnover and detrends by the firm’s own 180-trading-day median, scaling Nasdaq volume down by 38% after 1997 and by 50% before, following Anderson and Dyl (2005), who document that dealer participation roughly doubles reported Nasdaq

volume (a 38% reduction in their 1997–2002 sample; practitioners use about 50% before). The paper’s data takes the last day’s raw turnover minus its own 180-day mean, with no market adjustment and no scaling. **Settled: FNW’s definition, averaged over the days of the month: daily market-adjusted turnover minus its trailing 180-trading-day median, Nasdaq volume scaled as in Anderson and Dyl.** Garfinkel’s own detrending window is an event-study control period and has no monthly analogue; FNW’s 180-day median is the anomaly-literature version and the one the paper cites.

Turnover (TNOVR). Datar, Naik and Radcliffe (1998) measure turnover as the average number of shares traded over the preceding three months divided by shares outstanding as of the period, excluding stocks whose share count changed in those months. FNW’s characteristic is “last month’s volume over shares outstanding”: the one-month version, with the volume lagged relative to the share count exactly as in Datar, Naik and Radcliffe. The paper’s data is last month’s volume over this month’s shares, which *is* FNW’s definition; Table A.1 dropped the words “last month’s”. **Settled: no change.** The earlier classification of this signal as a documentation discrepancy is withdrawn; it was a misreading of the condensed appendix.

Industry-adjusted book-to-market, size, profit margin and asset turnover (aBEME, aSIZE, aPM, aSAT). FNW: the characteristic minus its average at the Fama-French 48 industry level, citing Asness, Porter and Stevens (2000) for book-to-market and size and Soliman (2008) for profit margin and asset turnover. Asness, Porter and Stevens (2000, section 2) assign each stock every month to one of the 48 industries of Fama and French (1997) by its four-digit SIC code, define the within-industry measure as the firm’s characteristic minus the *equally weighted* average of the characteristic over the industry’s constituents, and require at least three firm-years in an industry. Soliman (2008) defines profit margin as operating income over sales and asset turnover as sales over *average* net operating assets, and does not industry-adjust either in the published paper (the adjustment belongs to his 2004 dissertation); the industry adjustment of PM and SAT is therefore FNW’s construction, built the same way as Asness, Porter and Stevens’s. The paper’s data demeans within 27 industry groups of unknown origin, recovered from its panels. **Settled: the Fama-French 48 industries by SIC range (Ken French’s list), the equal-weighted industry mean, at least three firms, CRSP’s name-history SIC code, for all four.** FNW’s “SAT” is sales over total assets rather than Soliman’s turnover over net operating assets; the paper’s aSAT adjusts SAT and that is kept.

aPM. Same construction; in addition its cell means cannot be rebuilt (they were computed over a wider firm set than the panels hold) and its long-short alpha is 0.00012 a month, so the long leg is not identified. **Settled: dropped from the site’s portfolios; the aPM sort is not a signal in any meaningful sense.**

Conventions kept. The panels revealed several implementation choices that the sources leave open and that do not change what a signal measures: the beta window (the firm’s last 249 trading days), zero-volume days excluded from the volume dispersion, $n - 1$ in the residual standard deviation, the day-level range-or-quote spread, dividends in dollars over market capitalisation. These are documented in `signals.pdf` and kept.

Summary table

Signal	Original source	Paper's data	Site
OA, AOA	Sloan (1996): minus depreciation	plus depreciation	Sloan, lagged assets
SUV	Garfinkel (2009): daily standardised residual, prior-window regression	last day of a within-month regression	daily residuals from month $t - 1$'s fit, mean over month t
DTO	Garfinkel (2009): daily market-adjusted turnover minus control-period mean; FNW: minus 180-day median, Nasdaq scaled	last day's raw turnover minus 180-day mean, no market adjustment	FNW's, mean over the month
TNOVR	Datar et al. (1998): 3-month average volume over current shares; FNW: last month's	last month's volume over current shares	unchanged (= FNW)
aBEME, aSIZE	Asness et al. (2000): FF48, equal-weighted mean, three firms	27 recovered groups	FF48 equal-weighted mean
aSAT	Soliman (2008) defines SAT's cousin ATO and does not adjust; FNW: FF48 mean	27 recovered groups	FF48 equal-weighted mean
aPM	Soliman (2008)	not reproducible; alpha zero	dropped

The options

1. **As the paper did it.** The site's portfolios are the paper's. Table 1 reproduces row by row and the reference series and the headline series share one set of portfolios. Six portfolios do not mean what their names say: the accruals sort is partly a depreciation sort; the SUV and DTO sorts are one-day signals sampled on the last trading day.
2. **As the paper describes it.** The six are rebuilt from the documented definitions and aPM is dropped. Portfolios mean what they say and the site can cite the definitions. Six rows of the portfolio table differ from the JFE numbers and this has to be stated; the mapping loses one regressor.
3. **Signal by signal.** For instance the accruals and industry fixes with SUV and DTO kept as published, on the argument that those two carry small wedges either way (-5 and -6 points on the long leg).

What the evidence says about the size of the choice

The firm-level series is affected modestly: six of 57 regressors and six of 570 portfolios per weighting. The portfolio-level table is affected exactly in those six rows. The conventions the panels revealed

that do *not* change what a signal measures are kept under every option: the 249-trading-day window for the daily beta, zero-volume days excluded from the volume dispersion, $n - 1$ in the idiosyncratic volatility, the day-level spread, dividends in dollars over market capitalisation.

The decision and its reasons

Option 2, with option 1 run alongside as the labelled reference; the definitions are the ones the source review above settles, signal by signal. The principle behind it, in Christian’s words, is that the point of the anomalies is to use the existing specifications, not new ones that arise from an implementation mistake. The reason is the same as in decision A: the site should be able to state what its inputs are in one sentence, and “as defined in the paper’s appendix” is a sentence, while “as in the paper’s data, including a depreciation term with the wrong sign and two signals sampled on one day” is not one the site should have to make. The cost, six rows that differ from the published table, is disclosed rather than hidden.

4 Decision C: which cohorts the wedges are estimated on (open; default: the full available sample)

The options

The paper’s window (formations July 1964 to December 2002, cash flows to 2017; 463 cohorts), the full available sample (formations to January 2011, cash flows to December 2025; 559 cohorts), or a later window.

What the evidence says

`stability.py` re-estimates all 570 wedges on subsamples of the cohorts, re-solving λ on each so the market wedge is zero there. Extending the paper’s window to the full sample changes almost nothing: correlation 0.99 between the two cross-sections, one long-short sign change, λ from 3.97 to 3.53 on the same data. Dropping the early cohorts is different. The two halves (1964–87 and 1988–2011) correlate 0.45 where two halves of a constant cross-section would correlate about 0.68 given their noise; the value ratios reverse sign between halves, profitability spreads shrink, momentum grows; the 1980–95 third is nearly flat and unrelated to the rest. Any window starting in 1980 or later correlates 0.5 to 0.8 with the full cross-section, rests on one or two fifteen-year cycles (λ swings from 5.6 to 2.5 to 2.8 across the thirds), and moves the typical firm’s wedge by 10 to 20 points. The firm ranking is robust to the choice (correlations 0.9 and above); the level is not.

Default, pending the decision

The full available sample as the headline, with the two half-sample estimates shown as a range next to each firm’s number. A later window is not a better estimate of the same quantity; it is a different regime, and the site should show that regime dependence rather than pick a side.

5 What the site shows, given A and the defaults for B and C

- **Headline:** the candidate-3 firm wedge, full sample, documented signal definitions, with its half-sample range and an extrapolation flag for firms whose characteristics lie outside the profiles' support.
- **Reference:** the published construction (candidate 1, paper's definitions, paper's window), labelled "as published in the JFE".
- **Method page:** the definition of the wedge, the calibration (value-weighted market wedge zero), the weighting choices and the sentence each number can claim, the acceptance tests and their latest values.
- **Uncertainty:** the common component of the standard error (about 30 points, the market's realised path, the same for every firm) shown once as the uncertainty of the whole level; the firm-specific part (about 3 to 4 points relative, and the half-sample range of about ± 12) shown per firm.

6 Acceptance gates for every build

A series is published only if all of these hold on the latest run:

1. Replication: the "as published" spec verifies at least 56 of 57 characteristics against the paper's panels and Table 1 within a point on both legs.
2. Level test: slope of at least 0.85 and intercept within 3 points on equal-weighted deciles; slope of at least 0.75 on value-weighted deciles; the decile table reported.
3. Aggregate check: capitalisation-weighted mean of firm wedges within a point or two of the market wedge.
4. Out-of-sample fit: leave-one-family-out strict R^2 of the mapping reported and not below the previous build's by more than noise.
5. Ex-post returns: the fitted-wedge deciles' subsequent returns monotone at five years.

Appendix: scripts and files

build_panel.py	the 57 characteristics from CRSP and Compustat (--end 2025-12-31 --suffix _today for the current panel)
validate.py,	replication against Table 1 and against the paper's own
compare_panels.py	firm-month panels
portfolio_wedges.py	sorts, calibration, wedges; --weighting equal for the equal-weighted deciles
portfolio_profiles.py	profiles; --weighting equal for equal-weighted means
firm_wedges.py,	the mapping and the portfolio-level criteria
mapping_quality.py	
realised_levels.py	the realised-wedge level test and the aggregate check
stability.py	wedges on cohort subsamples
doc/quality.pdf,	the evidence behind every number here
doc/signals.pdf	
